

SPORT HMI for Audi

There are some indications of Sport_HMI app for Audi MIB2 firmware. But no-one yet have enabled it. Let's check methods from the VW/Skoda/Seat units and let's check how the [LSD.JXE](#) in Audi looks like when it comes to references to sport modes.



Audi MMI SportHMI mockup

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VW/SEAT/Skoda SportHMI enabling methods

For whatever reason, VW/SEAT/Skoda vehicles support SPORT HMI out from the factory, without some deep modifications of the system. All it needs is adaptation and FEC/SWaP code.

Details: [SPORT HMI Performance Monitor](#)

GUI assets in Audi firmware

There's a file `SPORT_HMI.kzb` - a .kzb package. After extracting, it looks like Audi firmware contains some GUI assets for Sport Display submenu. There's clock gauge for power and torque, as well as bar gauge for temperature and boost. Also rendered fonts for the large readouts inside the clock gauges.

So if we are looking for needed textures - all is here.

Here we can see that those textures are referenced in `LSD.JXE` classes:

```
public static String SPORT_HMI = "SPORT_HMI.kzb";
```

```
public static int mib2_sport_hmi_kzbFile = 100;
```

SportHMI adaptation channels

We all know that there are adaptation channels related to the Performance Monitor in VW, SEAT, and Skoda → http://wiki.mib-helper.com/index.php?title=SPORT_HMI_Performance_Monitor#Adaptation



Audi MIB2 MMI SportHMI GUI assets

And it looks like in Audi those are also connected to Sport Display function. Look at this part of code:

```
if (this.isComponentAvailable((short)52, carFuncAdap)) {
    this.addCarComponent(new SportComponentEvo(this));
    this.addCarComponent(new SportKombiComponentEvo(this));
}
```

The `isComponentAvailable((short)52, carFuncAdap)` simply checks if 52nd channel in `Car_Function_Adaptations_Gen2` section is set to on. And 52nd channel is... You know which one - `[L0]_menu_display_SportHMI`.

Sport and Sport Kombi classes

It looks like the app logic is split into two separate classes:

- `SportHMI`
- `SportHMI-Kombi`

`Kombi` is designed for instrument cluster data ("Kombiinstrument"), which is the oil temperature.

Here's a statement that creates the screen:

```
protected Screen createScreen(int n, int n2) {
    switch (n) {
        (...)
        case 600289: {
            return CarScreenBag1.CARSPORTMAIN(this, n2);
        }
    }
}
```

This one returns name of the screen for diagnostic purposes

```
case 600289: {
    return "CAR_SPORT_MAIN";
}
```

And this one executes conditions

```
@Override
public void executeCondition(int n, int n2, HMIView[] hMIViewArray, int n3) {
    switch (n2) {
        (...)
        case 600289: {
            this.executeConditioncARSPORTMAINSscreen(n, hMIViewArray, n3);
            break;
        }
    }
}
```

...which look like this:

```
private void executeConditioncARSPORTMAINSscreen(int n, HMIView[] hMIViewArray, int n2) {
    switch (n) {
        case 601056: {
            if (hMIViewArray[0] == null) break;
            ((ScreenWidgetEVO)hMIViewArray[0]).setOpenSelectionDrawerByHkReturn(!
this.evaluateSimpleChoiceModelValueEqualsCondition(-534050560, n2, -214497024));
            break;
        }
        case 602113: {
            if (hMIViewArray[0] != null) {

((ContainerController)hMIViewArray[0]).setVisible(this.evaluateSimpleChoiceModelValueEqualsCondition(1992
5248, n2, 1));
            }
            if (hMIViewArray[1] == null) break;

((OilTemperatureGaugeController)hMIViewArray[1]).setVisible(this.evaluateSimpleChoiceModelValueEqualsCond
ition(19925248, n2, 1));
            break;
        }
        case 602114: {
            if (hMIViewArray[0] == null) break;

((ChargingAirPressureGaugeController)hMIViewArray[0]).setVisible(this.evaluateSimpleChoiceModelValueEqual
sCondition(36702464, n2, 1));
            break;
        }
        case 602116: {
            if (hMIViewArray[0] == null) break;

((ContainerController)hMIViewArray[0]).setVisible(this.evaluateSimpleChoiceModelValueEqualsCondition(7025
6896, n2, 1));
            break;
        }
        case 602118: {
            if (hMIViewArray[0] == null) break;

((ContainerController)hMIViewArray[0]).setVisible(this.evaluateSimpleChoiceModelValueEqualsCondition(1038
11328, n2, 1));
            break;
        }
    }
}
}
```

From the above:

- 602113 - Oil temperature
- 602114 - Boost pressure
- 602116 - *TODO*
- 602118 - *TODO*

State machine class is also referring to the SportHMI:

```
public static final int LD_CAR_SEL_SPORTDISPLAY;
```

Meaning of the above would be "Sport Display was selected from the CAR menu Left Drawer".

- 601 - SPORT_POWER
- 602 - SPORT_POWER_MAX
- 603 - SPORT_POWER_CURRENT
- 604 - SPORT_TORQUE
- 605 - SPORT_TORQUE_MAX
- 606 - SPORT_TORQUE_CURRENT
- 607 - SPORT_OIL_TEMPERATURE
- 608 - SPORT_AIR_PRESSURE

Why no Sport submenu in Car menu?

It looks like everything is there. All the programming logic, adaptation to activate, data pulls, GUI assets, inits... So why it is not working?

One of my theories is that at some point of development they decided to move the Sport HMI functionality from MIB to VC. On one side it makes sense - feature will not be duplicated on both displays. But on the other hand VC is optional equipment and not all cars have it.

For whatever reasons it was - they needed to remove Sport HMI from Audi MIB. And by "remove" I mean "make it not visible for user". That's why all the mentioned above parts of code and assets are there. So how they did it? And how to restore it?

Maybe let's analyse how the MENU and CAR MENU structures looks like:

- Car
 - Audi Drive Select
 - Charging (eTron)
 - Statistics (eTron)
 - Vehicle Settings
 - Drive Assist
 - Air Condition
 - Service and Checks
 - Boardbook
- Sound
- Media
- Phone
- ...

By the first look, I can see some similarities to the `CarEvoMenuEntryStructure.java` class:

```

MenuEntry menuEntry2 = this.entryFactory.createMenuEntry(1596459264, "CAR_FUNC_CHARISMA", -282457856);
MenuEntry menuEntry3 = this.entryFactory.createMenuEntry(1630013696, "CAR_FUNC_SETTINGS", -1305802496);
MenuEntry menuEntry4 = this.entryFactory.createMenuEntry(1646790912, "CAR_FUNC_DRIVE_ASSIST",
-1339356928);
MenuEntry menuEntry5 = this.entryFactory.createMenuEntry(1663568128, "CAR_FUNC_AIR_CONDITION",
-1389688576);
MenuEntry menuEntry6 = this.entryFactory.createMenuEntry(1680345344, "CAR_FUNC_AUXHEAT", -1356134144);
MenuEntry menuEntry7 = this.entryFactory.createMenuEntry(10, "CAR_FUNC_AUX_AC", -1372911360);
MenuEntry menuEntry8 = this.entryFactory.createMenuEntry(11, "CAR_FUNC_AUX_COMBINED", 1210976512);
MenuEntry menuEntry9 = this.entryFactory.createMenuEntry(1713899776, "CAR_FUNC_SERVICE", -1322579712);
MenuEntry menuEntry10 = this.entryFactory.createMenuEntry(1177094400, "CAR_FUNC_BORDBOOK", 1412172032);
MenuEntry menuEntry11 = this.entryFactory.createMenuEntry(640289024, "CAR_FUNC_CHARGE", -1842607872);
MenuEntry menuEntry12 = this.entryFactory.createMenuEntry(-383121152, "CAR_FUNC_STATISTICS", -349370112);
MenuEntry menuEntry13 = this.entryFactory.createMenuEntry(12, "CAR_FUNC_SPORT", 120588544);

```

Now, let's check how the top-level menus are initialised and compare them to the SportHMI class to find what's missing. Maybe this will point us into the right direction.

After quick check and compare, it looks like there's one function missing from the SportComponentEvo class. It's `updateMenuEntryVisibility()` that appears in other car submenu classes. Not only those top-level. All of them. Is this it?

Moment of Doubt

Had a chance to test-drive an Audi SQ7 (4M) and played a bit with the Drive Select functions. There are "off-road" and "allroad" modes which enable additional element - Tilt Angle Display (TAD). This new element is visible directly on the Drive Select screen, and not as a separate submenu in the CAR menu.

So it got me wondering. What if the SportHMI is also just an additional widget that should be displayed on the Drive Select screen when "Sport" or "Race" mode is selected? If so, it would mean that this function might be controlled via Gateway with proper dataset. Or it's a separate submenu but still enabled with a gateway dataset?

I need to look into the Charisma menu and TAD elements in LSH.JXE...

```

private IMenuEntry[] buildMenuCharisma(LogChannel logChannel) {
    // (...)
    menuEntry8.setFunctionalStateValues(new int[]{4}); //LIFT
    menuEntry2.setFunctionalStateValues(new int[]{4}); //OFFROAD
    menuEntry10.setFunctionalStateValues(new int[]{4}); //LIFT OFFROAD
    menuEntry5.setFunctionalStateValues(new int[]{8}); //EFFICIENCY
    // (...)
    MenuEntry menuEntry11 = this.entryFactory.createMenuEntry(250, "CHARISMA_EFFICIENCY_BCME_ADD_INFO",
1898776832);
    MenuEntry menuEntry12 = this.entryFactory.createMenuEntry(252,
"CHARISMA_EFFICIENCY_BCME_CONSUMPTION_BARGRAPH", 1613564160);
    MenuEntry menuEntry13 = this.entryFactory.createMenuEntry(251, "CHARISMA_EFFICIENCY_BCME_CONSUMER_LIST",
1915554048);
    menuEntry11.setChildren(new IMenuEntry[]{menuEntry12, menuEntry13});
    MenuEntry menuEntry14 = this.entryFactory.createMenuEntry(326, "TAD_MAIN", 1999374592);
    MenuEntry menuEntry15 = this.entryFactory.createMenuEntry(328, "TAD_PITCH_ANGLE", 1965820160);
    MenuEntry menuEntry16 = this.entryFactory.createMenuEntry(327, "TAD_ROLL_ANGLE", 1982597376);
    menuEntry14.setChildren(new IMenuEntry[]{menuEntry15, menuEntry16});
    return new IMenuEntry[]{menuEntry, menuEntry2, menuEntry3, menuEntry4, menuEntry5, menuEntry7,
menuEntry6, menuEntry8, menuEntry10, menuEntry9};
}

```

Nope! It looks like only Efficiency Mode and Lift/Offroad Modes have children elements. No SportHMI mentions here. And previously mentioned `LD_CAR_SEL_SPORTDISPLAY` was linked to left drawer,

which means that the SportHMI needs to be present in the CAR menu.

But there's a RACE Mode mentioned here and there.

```
MenuEntry menuEntry9 = this.entryFactory.createMenuEntry(238, "CHARISMA_PROFILE_RACE", 305203456);
```

```
public boolean isRacesport() {
    return this.racesport;
}
```

Maybe that's it? Maybe the RACE Charisma mode needs to be enabled from the SportHMI to be accessible?

Just to be sure, I will check if there are some "RACE" references in the code. But not now.

Figuring out the missing data

Played around with KeyCombo (that's a separate topic) and found that Audi HMI has an build-in app for HMI tests. It shows a lot of data about visible items and data provided by the rest of the car.

Here we can see which information is provided **state: 2**, and which is not **state: 0**:

```
SportHMI
DynamicVehicleInfoHighFrequentViewOptions(
    longitudinalAcceleration=ViewOption(state: 2, reason: 0),
    engineSpeed=ViewOption(state: 2, reason: 0),
    powermeter=ViewOption(state: 0, reason: 0),
    lateralAcceleration=ViewOption(state: 2, reason: 0),
    relChargingAirPressure=ViewOption(state: 2, reason: 0),
    absChargingAirPressure=ViewOption(state: 2, reason: 0),
    currentOutputPower=ViewOption(state: 2, reason: 0),
    currentTorque=ViewOption(state: 2, reason: 0),
    yawRate=ViewOption(state: 2, reason: 0),
    brakePressure=ViewOption(state: 2, reason: 0),
    wheelAngle=ViewOption(state: 2, reason: 0),
    acceleratorPosition=ViewOption(state: 2, reason: 0),
    vehicleSpeed=ViewOption(state: 2, reason: 0),
    systemCycleDistance=ViewOption(state: 2, reason: 0),
    torqueDisteributionFrontLeft=ViewOption(state: 0, reason: 0),
    torqueDisteributionFrontRight=ViewOption(state: 0, reason: 0),
    torqueDisteributionRearLeft=ViewOption(state: 0, reason: 0),
    torqueDisteributionRearRight=ViewOption(state: 0, reason: 0),
    realVehicleSpeed=ViewOption(state: 2, reason: 0),
    shiftLight=ViewOption(state: 0, reason: 0),
    allWheelDriveTorque=ViewOption(state: 0, reason: 0),
    espOversteering=ViewOption(state: 0, reason: 0),
    espUndersteering=ViewOption(state: 0, reason: 0),
    espWheelVelovityFrontLeft=ViewOption(state: 0, reason: 0),
    espWheelVelovityFrontRight=ViewOption(state: 0, reason: 0),
    espWheelVelovityRearLeft=ViewOption(state: 0, reason: 0),
    espWheelVelovityRearRight=ViewOption(state: 0, reason: 0),
    relAllWheelDriveTorque=ViewOption(state: 0, reason: 0)
)
SemiStaticDataViewOptions(
    maxChargingAirPressure=ViewOption(state: 2, reason: 0),
    maxOutputPower=ViewOption(state: 2, reason: 0),
    maxTorque=ViewOption(state: 0, reason: 0),
    engineDisplacement=ViewOption(state: 2, reason: 0),
    typeOfInjectionSystem=ViewOption(state: 2, reason: 0)
)
```

And here we can check which components in the SPORT HMI are fully functional and which are not:

```
CAR_FUNC_SPORT
  State: FUNCTIONAL
  ModelID: 602119
  Parent: CAR_MAIN
  Children:
    SPORT_POWER
    SPORT_TORQUE
    SPORT_OIL_TEMPERATURE
    SPORT_AIR_PRESSURE

SPORT_POWER
  State: FUNCTIONAL
  ModelID: 602116
  Parent: CAR_FUNC_SPORT
  Children:
    SPORT_POWER_MAX
    SPORT_POWER_CURRENT

SPORT_POWER_MAX
  State: FUNCTIONAL
  ModelID: -1
  Parent: SPORT_POWER
  Children: none

SPORT_POWER_CURRENT
  State: FUNCTIONAL
  ModelID: -1
  Parent: SPORT_POWER
  Children: none

SPORT_TORQUE
  State: FUNCTIONAL
  ModelID: 602118
  Parent: CAR_FUNC_SPORT
  Children:
    SPORT_TORQUE_MAX
    SPORT_TORQUE_CURRENT

SPORT_TORQUE_MAX
  State: INVISIBLE
  ModelID: -1
  Parent: SPORT_TORQUE
  Children: none

SPORT_OIL_TEMPERATURE
  State: Invisible
  ModelID: 602113
  Parent: CAR_FUNC_SPORT
  Children: none

SPORT_AIR_PRESSURE
  State: FUNCTIONAL
  ModelID: 602114
  Parent: CAR_FUNC_SPORT
  Children: none
```

Two things (`SPORT_TORQUE_MAX` and `SPORT_OIL_TEMPERATURE` are invisible. This partially correlates with unavailable `maxTorque`.

Is it that this missing data is what's locking the SPORT submenu to appear? Let's find out!

Logging HMI

The `run_lsd.sh` script, that starts the `LSD.jxe` can be adjusted for logging different components.

App.Car.Sport=5

Let's start with **App.Car.Sport** set to level 5, so it will output some data.

```
updateBCViewOptions:BCViewOptions(
  currentConsumption1=ViewOption(state: 0,reason: 0),
  currentConsumption2=ViewOption(state: 0,reason: 0),
  currentRange1=ViewOption(state: 2,reason: 0),
  currentRange2=ViewOption(state: 0,reason: 0),
  totalDistance=ViewOption(state: 2,reason: 0),
  shortTermAverageConsumption1=ViewOption(state: 2,reason: 0),
  shortTermAverageConsumption2=ViewOption(state: 0,reason: 0),
  shortTermGeneral=ViewOption(state: 2,reason: 0),
  longTermAverageConsumption1=ViewOption(state: 2,reason: 0),
  longTermAverageConsumption2=ViewOption(state: 0,reason: 0),
  longTermGeneral=ViewOption(state: 2,reason: 0),
  cycleAverageConsumption1=ViewOption(state: 0,reason: 0),
  cycleAverageConsumption2=ViewOption(state: 0,reason: 0),
  cycleGeneral=ViewOption(state: 0,reason: 0),
  vZADisplay=ViewOption(state: 2,reason: 0),
  lifeTipsDisplay=ViewOption(state: 2,reason: 0),
  consumerDisplay=ViewOption(state: 2,reason: 0),
  tankLevel1=ViewOption(state: 2,reason: 0),
  tankLevel2=ViewOption(state: 0,reason: 0),
  refuelVolumel=ViewOption(state: 0,reason: 0),
  refuelVolume2=ViewOption(state: 0,reason: 0),
  menue1Config=ViewOption(state: 0,reason: 0),
  menue2Config=ViewOption(state: 0,reason: 0),
  menue3Config=ViewOption(state: 0,reason: 0),
  resetMenue1=ViewOption(state: 2,reason: 0),
  resetMenue2=ViewOption(state: 2,reason: 0),
  resetMenue3=ViewOption(state: 0,reason: 0),
  oilTemperature=ViewOption(state: 0,reason: 0),
  digitalSpeed=ViewOption(state: 0,reason: 0),
  stopwatch=ViewOption(state: 0,reason: 0),
  vzaMFA=ViewOption(state: 0,reason: 0),
  speedWarning=ViewOption(state: 2,reason: 0),
  gearRecommendation=ViewOption(state: 0,reason: 0),
  rearSeatbeltWarning=ViewOption(state: 0,reason: 0),
  indications=ViewOption(state: 0,reason: 0),
  outsideTemperature=ViewOption(state: 0,reason: 0),
  bcSetFactoryDefault=ViewOption(state: 2,reason: 0),
  vehicleStateList=ViewOption(state: 0,reason: 0),
  statisticsTimeAC1=ViewOption(state: 0,reason: 0),
  statisticsTimeAC2=ViewOption(state: 0,reason: 0),
  statisticsTimeRE=ViewOption(state: 0,reason: 0),
  statisticsTimeZE=ViewOption(state: 0,reason: 0),
  statisticsTimeCurrentPeriodAC1=ViewOption(state: 0,reason: 0),
  statisticsTimeCurrentPeriodAC2=ViewOption(state: 0,reason: 0),
  statisticsTimeCurrentPeriodRE=ViewOption(state: 0,reason: 0),
  statisticsTimeCurrentPeriodZE=ViewOption(state: 0,reason: 0),
  statisticsDistanceAC1=ViewOption(state: 0,reason: 0),
  statisticsDistanceAC2=ViewOption(state: 0,reason: 0),
  statisticsDistanceRE=ViewOption(state: 0,reason: 0),
  statisticsDistanceZE=ViewOption(state: 0,reason: 0),
  statisticsDistanceCurrentIntervallAC1=ViewOption(state: 0,reason: 0),
  statisticsDistanceCurrentIntervallAC2=ViewOption(state: 0,reason: 0),
  statisticsDistanceCurrentIntervallRE=ViewOption(state: 0,reason: 0),
  statisticsDistanceCurrentIntervallZE=ViewOption(state: 0,reason: 0),
  statisticsReset=ViewOption(state: 0,reason: 0),
  statisticsConfig=ViewOption(state: 0,reason: 0),
  oilTemperatureValue=ViewOption(state: 0,reason: 0), // ← ← ← ← ← ← ← ← NOT OK
  coolantTemperature=ViewOption(state: 0,reason: 0),
  comfortPowerConsumptionST=ViewOption(state: 0,reason: 0),
  statisticDistanceEUkm=ViewOption(state: 0,reason: 0),
  statisticDistanceEUmls=ViewOption(state: 0,reason: 0),
  totalCurrentRange=ViewOption(state: 0,reason: 0),
  resetMaxValues=ViewOption(state: 0,reason: 0),
  zeroEmissionDistanceST=ViewOption(state: 0,reason: 0),
  zeroEmissionTimeST=ViewOption(state: 0,reason: 0),
  zeroEmissionDistanceLT=ViewOption(state: 0,reason: 0),
  zeroEmissionTimeLT=ViewOption(state: 0,reason: 0),
  zeroEmissionDistanceCY=ViewOption(state: 0,reason: 0),
  zeroEmissionTimeCY=ViewOption(state: 0,reason: 0),
  maxValues=ViewOption(state: 0,reason: 0),
```

```

resetTimeStampST=ViewOption(state: 0,reason: 0),
resetTimeStampLT=ViewOption(state: 0,reason: 0),
resetTimeStampCY=ViewOption(state: 0,reason: 0),
comfortPowerConsumptionLT=ViewOption(state: 0,reason: 0),
comfortPowerConsumptionCY=ViewOption(state: 0,reason: 0),
astaMFA=ViewOption(state: 0,reason: 0),
resetTrip=ViewOption(state: 0,reason: 0),
configuration=BCConfiguration(
    primaryEngineType=7,
    secondaryEngineType=0,
    fisAdditionalConfiguration=BCFISAdditionalConfiguration(
        fis1=true,
        fis2=true,
        fis3=false,
        stopWatch=true,
        oilTemp=false,
        digitalSpeed=true,
        refuelVolume=false,
        speedWarning=true,
        coolantTemp=false,
        secondarySpeed=false,
        vza=true,
        resetTime=false,
        comfortPowerConsumption=false,
        zeroEmissionTime=false,
        zeroEmissionDistance=false,
        vzaMfa=false,
        bcmeConsumerList=false,
        bcmeLiveTips=false,
        astaMfa=false
    ),
    transmittableElementsVehicleState=BCTransmittableElements(
        ra0=1,ra1=7,ra2=1,ra3=7,raF=88
    ),functionSupport=BCFunctionSupport(
        resetTrip=false
    )
),
valid:1

```

```

00:00:42.720 INFO <App.Car.Sport>
updateBCOilTemperatureValue:null,
valid:2 // ← ← ← ← ← ← ← NOT OK

```

```

00:00:42.715 INFO <App.Car.Sport>
updateSemiStaticVehicleData:SemiStaticVehicleData(
    maxChargingAirPressure=2.5,
    maxOutputPower=228,
    maxTorque=0, // ← ← ← ← ← ← ← NOT OK
    engineDisplacement=2.0,
    typeOfInjectionSystem=0),
valid:1

```

```

00:00:42.722 INFO <App.Car.Sport>
updateDynamicVehicleInfoHighFrequent:DynamicVehicleInfoHighFrequent(
    longitudinalAcceleration=0.0,
    engineSpeed=699,
    powermeter=0,
    lateralAcceleration=-0.01,
    relChargingAirPressure=0,
    absChargingAirPressure=0.29,
    currentOutputPower=1.8,
    yawRate=-0.09,
    brakePressure=0.0,
    wheelAngle=8,
    acceleratorPosition=0,
    vehicleSpeed=CarBCSpeed(
        speedValueState=1,
        speedValue=0.0,
        speedUnit=0),
    currentTorque=25,
    systemCycleDistance=0,
    torqueDistributionFrontLeft=0,
    torqueDistributionFrontRight=0,
    torqueDistributionRearLeft=0,

```

```

torqueDistributionRearRight=0,
realVehicleSpeed=CarBCSpeed(
  speedValueState=1,
  speedValue=0.0,
  speedUnit=0),
shiftLight=0,
allWheelDriveTorque=0,
espOversteering=0.0,
espUndersteering=0.0,
espWheelVelocityFrontLeft=0.0,
espWheelVelocityFrontRight=0.0,
espWheelVelocityRearLeft=0.0,
espWheelVelocityRearRight=0.0,
relAllWheelDriveTorque=0.0),
valid:1

```

This proves two things: 1. Sport_HMI exists, works, creates output, and should be functional. 2. Some of the data from DSI is not provided or is zeroed.

App.Car.MER=5

Since the Sport_HMI is clearly there, but it's not visible in the CAR menu, let's check how the Menu Entry Register logs look like.

```

00:00:11.913 INFO <App.Car.MER> [MenuEntry('CAR_MAIN')#setChildren] children=
CAR_FUNC_CHARISMA
CAR_FUNC_SETTINGS
CAR_FUNC_DRIVE_ASSIST
CAR_FUNC_AIR_CONDITION
CAR_FUNC_AUXHEAT
CAR_FUNC_AUX_AC)
CAR_FUNC_AUX_COMBINED)
CAR_FUNC_SERVICE
CAR_FUNC_BORDBOOK
CAR_FUNC_CHARGE
CAR_FUNC_STATISTICS
CAR_FUNC_SPORT

00:00:12.628 INFO <App.Car.MER> [MenuEntry('SPORT_TORQUE')#setChildren] children=
SPORT_TORQUE_MAX
SPORT_TORQUE_CURRENT

00:00:12.764 INFO <App.Car.MER> [MenuEntry('SPORT_POWER')#setChildren] children=
SPORT_POWER_MAX
SPORT_POWER_CURRENT

00:00:12.765 INFO <App.Car.MER> [MenuEntry('CAR_FUNC_SPORT')#setChildren] children=
SPORT_POWER
SPORT_TORQUE
SPORT_OIL_TEMPERATURE
SPORT_AIR_PRESSURE

```

Menu structure is created correctly. Now let's dig some more and check final status of each SPORT element.

```

00:00:34.812 INFO <App.Car.MER> [MenuEntry('SPORT_TORQUE_MAX')#setState] state='1' // ← NOT OK
00:00:35.084 INFO <App.Car.MER> [MenuEntry('SPORT_POWER_MAX')#setState] state='0'
00:00:35.086 INFO <App.Car.MER> [MenuEntry('SPORT_OIL_TEMPERATURE')#setState] state='1' // ← NOT OK
00:00:35.224 INFO <App.Car.MER> [MenuEntry('SPORT_TORQUE')#setState] state='0'
00:00:35.224 INFO <App.Car.MER> [MenuEntry('SPORT_TORQUE_CURRENT')#setState] state='0'
00:00:35.500 INFO <App.Car.MER> [MenuEntry('SPORT_POWER')#updateState] state='0'
00:00:35.500 INFO <App.Car.MER> [MenuEntry('SPORT_POWER_CURRENT')#setState] state='0'
00:00:35.647 INFO <App.Car.MER> [MenuEntry('SPORT_AIR_PRESSURE')#setState] state='0'
00:00:35.649 INFO <App.Car.MER> [MenuEntry('CAR_FUNC_SPORT')#updateState] state='0'

```

It looks like the `state='0'` is reported if the component is working correctly. And two component's are not. `SPORT_TORQUE_MAX` and `SPORT_OIL_TEMPERATURE` - same two that were missing or zeroed in DSI-provided data. Huh...

How to enable it?

I have 4 ideas, actually. Not sure which is best and if any would work.

1. modify SportHMI classes
2. patch DSI with missing data
3. ~~deregister components from Menu Entry that cause it to be invisible~~
4. ~~create a shortcut for SportHMI to bypass the Menu Entry Registry all together~~
5. patch menu entry registry
6. patch carScreenBag

Modifications with custom data

//TODO ideas:

- oil temp → gearbox temp
- bar: acceleratorPosition=ViewOption(state: 2, reason: 0)
- bar: brakePressure=ViewOption(state: 2, reason: 0)

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